

Paper 17.50 - Error-Correction Tower Claim Contract

CQE-paper-17.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-17.50.md

Paper 17.50 - Error-Correction Tower Claim Contract

Purpose

This contract governs every Paper 17 claim. It admits verified code and root-shell receipts and rejects unproved promotion into Leech glue, Weyl-table, or physical error-correction claims.

Admitted Claims

Admit a claim when it provides:

```
rung number
code or lattice object
closed parameter receipt
minimum weight or norm where relevant
self-duality or orthogonality status where relevant
open glue or extractor boundary where relevant
```

Rejected Promotions

Reject by default:

```
Golay ingredients -> completed Leech lattice
E8 seed -> W(E8) lookup extractor
Niemeier registry -> semantic N-to-terminal map
code tower -> new physical error-correction theorem
root-shell profile -> exact glue cosets
```

Hidden-Guess Rule

All non-math diagnostic choices must be hidden until after classification. Allowed labels are:

```
closed code rung
closed powered bound
bounded root-shell profile
ingredient-only layer
open promotion
invalid overclaim
```

Linked Receipt

Current receipt:

```
production/formal-papers/CQE-paper-17/error_correction_tower_receipt.json
```

Current verdict:

```
bounded error-correction tower: pass
Leech glue action: open
W(E8) extractor: open
semantic N-to-terminal map: open
```

Boundary Failure

The contract fails if a later paper cites Paper 17 as proof of a completed Leech construction, a completed Weyl extractor, or a physical error-correction theorem.

The contract passes if a later paper says Paper 17 supplies the verified tower and explicitly carries the open promotions forward.

Conclusion

Paper 17.50 keeps the tower strong by refusing to let its useful verified rungs hide the remaining obligations.